AIRWEAVE

The Air Picture That Knows When It’s Wrong

Project description
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One shared picture. Multiple imperfect sensors. Honest uncertainty.

AIRWEAVE is a proposed synthetic-data research demonstrator for combining imperfect observations from several low-cost sensor nodes into one common operational picture. Rather than hiding disagreements or presenting every estimate as equally reliable, it makes data quality, uncertainty, and degraded operation visible to the user.

Project status: concept and prototype specification. The accompanying presentation includes illustrative demo metrics that must be replaced with measured benchmark results before external performance claims are made.

Executive summary

Distributed sensing systems rarely receive perfect, synchronized observations. Reports can be noisy, delayed, out of order, or missing entirely. These conditions can produce duplicate tracks, unstable identities, and stale estimates that appear current. AIRWEAVE is designed to make those failure modes observable and testable through a reproducible simulation, a fusion pipeline, and an evidence-first dashboard.

The project starts with synthetic moving objects and independent simulated sensors. A baseline estimator calculates position and velocity, carries uncertainty forward, and records which observations contributed to each estimate. Fault-injection controls allow a reviewer to introduce clock offsets, packet loss, sensor outages, and ambiguous observations while watching the operational picture respond.

1. Problem statement

A common operational picture is only useful when it represents both the best available estimate and the limitations of the evidence behind it. Cheap sensor nodes may differ in accuracy, field of view, update rate, clock quality, and communication reliability. Simply plotting every observation or merging them without robust time handling can create misleading output.

AIRWEAVE focuses on four connected problems:

- Noisy measurements: different sensors report inconsistent positions.
- Asynchronous reports: clock offsets and variable latency complicate comparison.
- Lossy communication: packets disappear and sensor nodes can go offline.
- Identity ambiguity: track identifiers can fragment or switch when objects are close together.

2. Proposed solution

AIRWEAVE is an uncertainty-aware sensor-fusion and evaluation platform. It combines synthetic observations into estimated tracks, reports uncertainty, preserves a record of evidence, and explicitly marks degraded or ambiguous conditions. The first implementation is centralized and deterministic to provide a testable reference. Distributed processing can be added after the core algorithms are validated.

Core capabilities

Sensor simulation. Configure independent nodes with noise, reporting rate, clock offset, packet loss, and outages.
Time and data quality. Validate observations, align timestamps, and flag stale reports.
State estimation. Estimate position and velocity with a documented Kalman-filter model and uncertainty.
Association awareness. Match observations conservatively and surface ambiguous associations.
Resilience testing. Inject faults and evaluate the effect on continuity and estimation quality.
Evidence-first dashboard. Show track state, contributing observations, sensor health, uncertainty, and event history.

3. Architecture

The proposed pipeline is: synthetic ground truth → independent sensor simulators → input validation and timestamp alignment → observation association and state estimation → track state, uncertainty and provenance → dashboard and benchmark evaluator.

Suggested stack: Python 3.11+, NumPy, SciPy, pytest, and pandas. A Streamlit interface is the quickest path to a functional demo. FastAPI and WebSockets or a React/TypeScript frontend can be introduced if the real-time presentation needs more customization. The core simulation and estimator should remain testable without the UI.

4. Real-world value and impact

AIRWEAVE is intended as an engineering validation tool, not a certified operational system. Its value is to help teams understand what happens when sensor quality or communications degrade, compare design choices with repeatable scenarios, and make uncertainty visible before it is mistaken for certainty.

Systems engineers: reproduce and diagnose failures.
Researchers: compare fusion and association methods against a known synthetic ground truth.
Integrators: measure the impact of different sensor qualities and communication settings.
Safety and assurance teams: inspect provenance, limitations, and degraded states.

Potential applications include robotics research, test-range experimentation, sensor-system validation, and educational demonstrations of distributed estimation. Real-world use would require representative sensor models, domain-specific validation, and appropriate safety and security reviews.

5. Evaluation and success criteria

The project should be judged on repeatable evidence, not on visual polish alone. Every benchmark run should record the scenario, random seed, number of observations, fault settings, and baseline definition.

Metric	Evaluation method
Position RMSE	Compare estimated position with independent simulated ground truth.
Velocity RMSE	Compare estimated velocity with ground-truth motion.
Identity continuity	Count ID switches, fragmentation, and track availability.
Uncertainty calibration	Check whether stated uncertainty matches observed estimation error.
Resilience	Repeat scenarios with increasing packet loss, clock offsets, and outages.
Efficiency	Measure update latency, CPU use, and memory consumption.

A single-sensor baseline should be compared with the fusion system under the same scenarios. Report cases where fusion does not improve results. Never hardcode benchmark values or imply that fusion always performs better.

6. Demonstration flow

1. Start a normal simulation with three sensor nodes.
2. Introduce a clock offset and observe how the system handles delayed reports.
3. Increase packet loss and disconnect a node.
4. Introduce two nearby synthetic objects to create ambiguous associations.
5. Reconnect the sensor and review track continuity, uncertainty, and event history.
6. Export benchmark results and compare against the single-sensor baseline.

7. Limitations and responsible use

Initial observations are synthetic and may not reproduce real sensor physics or operational environments. Estimator quality depends on assumptions, process and measurement noise, association rules, and the distribution of test cases. Covariance and confidence displays must be validated and calibrated. Simulation performance is not proof of field reliability.

AIRWEAVE is scoped to situational awareness, sensor-fusion research, and system testing. It does not implement weapon control, autonomous interception, engagement recommendations, or autonomous targeting. Human review and domain-specific validation remain necessary for any consequential use.

8. Deliverables

- Six-slide project presentation with a demo walkthrough.
- Project README and proposed modular architecture.
- Synthetic sensor simulator and repeatable fault scenarios.
- Tested fusion baseline and benchmark evaluator.
- Evidence-first dashboard and reproducible results report.

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